



Interactive Effects of Irrigation Scheduling and Sulphur Fertilization on Growth, Water use Efficiency, Yield and Oil Content of Indian Mustard (*Brassica juncea* L.) under Irrigated Conditions in Assam, India

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Authors' contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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Abstract

The experiment was carried out during the *rabi* season of 2019-20 at the Instructional-cum-Research Farm of Assam Agricultural University, Jorhat, to evaluate the effects of irrigation scheduling and sulphur fertilisation on growth parameters, crop growth rate (CGR), water use efficiency, yield, and oil content of Indian mustard (*Brassica juncea* L.) under irrigated conditions in Assam. The experiment was laid out in a split-plot design with two factors, viz., irrigation scheduling and sulphur fertilisation, each at different levels, with three replications. The treatments comprised six irrigation schedules, viz., I₀ - rainfed, I₁ - irrigation at 25-30 DAS, I₂ - irrigation at flowering (F), I₃ - irrigation at 25-30 DAS + flowering, I₄ - irrigation at 25-30 DAS + silique formation

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(SF), and I₅ - irrigation at 25-30 DAS + flowering + siliquae formation, and three sulphur levels, viz., S₀ - no application (control), S₁ - 20 kg ha⁻¹, and S₂ - 40 kg ha⁻¹. The results revealed that irrigation scheduling significantly influenced growth parameters, crop growth rate, water use, and yield attributes of Indian mustard. Three irrigations at 25-30 DAS + flowering + siliquae formation (I₅) and two irrigations at 25-30 DAS + flowering (I₃) or 25-30 DAS + siliquae formation (I₄) produced statistically at-par values for plant height at harvest, dry matter accumulation at 50 and 75 DAS, crop growth rate during 25-50 and 50-75 DAS, and branches plant⁻¹, and these values were higher than those recorded with one irrigation and rainfed conditions. The I₅ schedule resulted in higher evapotranspiration and total water use than I₃ and I₄, but recorded the lowest crop and field water use efficiencies. Plant height at 60 DAS and harvest, dry matter accumulation at 50 and 75 DAS, crop growth rate during 25-50 DAS, 50-75 DAS, and 75 DAS to harvest, and branches plant⁻¹ were highest with 40 kg S ha⁻¹, which was statistically at par with 20 kg S ha⁻¹ and significantly higher than the control. Although evapotranspiration and total water use varied little among sulphur levels, crop and field water use efficiencies were comparatively higher with 40 and 20 kg S ha⁻¹ than with the control.

Keywords: Indian mustard; irrigation scheduling; sulphur fertilization; crop growth rate; water use efficiency; evapotranspiration; irrigated conditions; rabi season.

1. Introduction

Oilseed crops play an important role in the Indian agricultural economy, with rapeseed and mustard being major oilseed crops widely cultivated across the country. In recent years, Indian mustard (*Brassica juncea* L.) has gained considerable importance in Assam as an alternative *rabi* oilseed crop because of its adaptability. However, because it is predominantly grown under rainfed conditions during the *rabi* season, productivity remains relatively low because of moisture stress and uneven rainfall distribution. Water deficit limits crop growth and adversely affects several physiological processes (Mohan *et al.*, 2025). Indian mustard responds strongly to irrigation, particularly during critical growth phases (Singh *et al.*, 2025). Timely irrigation has been reported to improve crop growth, yield attributes, and seed quality (Ray *et al.*, 2014). Therefore, appropriate irrigation at critical crop growth stages is essential for better growth and yield. Recent literature on Indian mustard likewise identifies drought and moisture scarcity as important constraints to crop productivity, reinforcing the relevance of efficient water management under water-limited production conditions (Baug & Darvhankar, 2026). Recent field evidence further indicates that irrigation scheduling, when integrated with nutrient management involving sulphur, can influence crop performance and quality-related responses in Indian mustard (Kumar *et al.*, 2025).

In addition to water availability, optimisation of nutrient supply plays an important role in increasing the productivity of Indian mustard. Rapeseed and mustard have a higher sulphur demand than most oilseed crops (Hegde & Babu, 2009). Rapeseed and mustard are highly sensitive to sulphur deficiency, which influences seed chemical composition and oil synthesis. Adequate sulphur availability also promotes seed and oil yield in *Brassica* crops and is essential for the synthesis of chlorophyll, proteins, enzymes, and vitamins (Shah *et al.*, 2022). As such, crop yield can be increased by up to 50% with an adequate supply of major nutrients, particularly sulphur. Sulphur application is therefore essential to meet the additional sulphur requirement of oilseed crops, particularly mustard, and to enhance oilseed production. Recent research has shown that sulphur-containing nutrient management can influence seed yield and associated crop responses in Indian mustard (Gupta *et al.*, 2024). Similarly, field evaluation of varying sulphur levels and application methods has demonstrated measurable differences in mustard productivity, supporting the agronomic importance of appropriate sulphur management (Mohapatra *et al.*, 2024).

In Assam, the *rabi* season (mid-October to February) is characterised by prolonged drought, with very low average rainfall of about 50 mm and high crop evapotranspiration demand. Given limited irrigation resources, identifying irrigation schedules for critical growth stages is necessary to maximise water productivity. Although irrigation and sulphur management have each been studied in mustard, their combined effects under the specific irrigated *rabi* conditions of Assam, particularly on growth, crop growth rate, water use efficiency, yield, and oil content, require assessment. Accordingly, the present study was based on the hypothesis that combining critical-stage irrigation scheduling with balanced NPK and sulphur fertilisation improves growth parameters, water use efficiency, and overall yield of Indian mustard under Assam conditions. Therefore, the objective of the present study was to evaluate the effects of irrigation scheduling and sulphur fertilisation on growth parameters, crop growth rate, water use efficiency, yield, and oil content of Indian mustard under irrigated conditions in Assam.

2. Materials and Methods

The present field experiment was conducted during the *rabi* season of 2019-20 at the Instructional-cum-Research Farm, Assam Agricultural University, Jorhat. The climate of Jorhat is subtropical and humid. The total rainfall during the crop growth period in *rabi* 2019-2020 (13 November 2019 to 22 February 2020) was 36.4 mm, received in 10 spells, of which five occurred during the crop maturity stage. The total evaporation during the crop-growing period was 123.7 mm, creating a rainfall deficit of 87.3 mm relative to transpirational demand. The weekly mean maximum temperature ranged from 20.7 °C to 29.7 °C, and the minimum temperature ranged from 8.4 °C to 18.1 °C during the experimental period. The weekly average morning relative humidity ranged from 97.1% to 100%, and bright sunshine ranged from 2.1 to 8.2 hours day⁻¹ during the experimental period. The average depth to the groundwater table was 88.35 cm and ranged from 58.6 to 121.2 cm below the soil surface.

The Indian mustard variety 'NRCHB101' was used in the experiment. The field experiment was laid out in a split-plot design with three replications. The main plots comprised six irrigation schedules (I₀ to I₅), while the subplots consisted of three sulphur levels, viz., S₀ (0 kg ha⁻¹), S₁ (20 kg ha⁻¹), and S₂ (40 kg ha⁻¹). Farmyard manure (FYM) @ 2 t ha⁻¹ and the recommended dose of N-P₂O₅-

K₂O @ 60-40-40 kg ha⁻¹ were applied uniformly to all plots. Nitrogen, phosphorus, and potassium were supplied through urea, diammonium phosphate (DAP), and muriate of potash (MOP), respectively. Sulphur was applied according to the treatments in the respective plots as sulphur microgranules containing 90 per cent sulphur; these were broadcast with the other fertilisers and incorporated into the soil. Irrigation was applied according to the treatments. The quantity of irrigation water was measured as described below.

The total amount of water applied to each plot was calculated by considering the area of each plot and the depth of irrigation, as follows:

$$q = a \times d$$

where,

$$\begin{aligned} q &= \text{quantity of water needed (m}^3\text{)} \\ a &= \text{area to be irrigated (3 m} \times \text{4 m = 12 m}^2\text{)} \\ d &= \text{depth of water (6 cm)} \\ \text{So, } q &= 12 \text{ m}^2 \times 0.06 \text{ m} \\ &= 0.72 \text{ m}^3 \\ &= 720 \text{ liters plot}^{-1} \end{aligned}$$

The discharge rate of the pump (Q) was determined using the simple container method as $Q = V/t$, where

V = volume of the container (15 litres) and

t = average time required for the pump discharge to fill the container (40 seconds)

Therefore, $Q = 15/40 = 0.38 \text{ litres sec}^{-1}$

Based on the discharge rate of the pump, each plot was irrigated for the time required to supply 720 litres plot⁻¹ according to the irrigation schedule, as follows:

Time (minutes) required to irrigate each plot (t) = $Q \times q/60$

$$= 0.375 \times 720/60 = 4.5 \text{ minutes}$$

Crop growth rate (CGR) is the increase in plant dry weight per unit area of crop over a given period. It was expressed in g m⁻² day⁻¹.

$$CGR = \frac{W_2 - W_1}{(t_2 - t_1) S}$$

where:

W₁ and W₂ are plant dry weights at times t₁ and t₂, respectively;
S is the land area over which dry matter was recorded.

Water use efficiency (WUE) of the crop under different treatments was calculated as follows:

$$WUE = \frac{\text{Weight of seed (Kg/ha)}}{\text{CU of crop (cm)}} \times 100$$

All data pertaining to the present investigation, wherever required, were statistically analysed for the split-plot design (SPD) using analysis of variance (ANOVA).

3. Results and Discussion

3.1 Growth Parameters and Crop Growth Rate (CGR)

Application of irrigation significantly influenced the various growth parameters, viz., plant height (Table 1), dry matter accumulation (Table 2), and branches plant⁻¹ (Table 3); however, plant population at harvest was non-significant. Crop growth rate (Table 4) at different crop growth stages was also significantly influenced by irrigation. Plant height at harvest, dry matter accumulation at 50 and 75 days after sowing (DAS), number of primary, secondary, and total branches plant⁻¹, and crop growth rate during 25-50 and 50-75 DAS were highest with three irrigations at 25-30 DAS, flowering, and siliquae formation stages (I₃), followed by two irrigations at 25-30 DAS + flowering (I₃) and 25-30 DAS + siliquae formation (I₄). These treatments were statistically at par and produced higher values than one irrigation (I₁ and I₂), whereas the lowest values were observed with no irrigation (I₀). Sharma *et al.* (2006) reported that increased irrigation might have contributed to cell turgidity and consequently enhanced the photosynthetic activity of the plant. This may be because increased irrigation enhanced nutrient uptake, translocation, and assimilation, thereby increasing plant height, number of leaves, and dry matter accumulation. Similar responses to two irrigations at 25 and 53 DAS (Roy *et al.*, 2017) and three irrigations at branching, flowering, and pod-filling stages (Yadav, 2005), or at 35, 60, and 90 DAS (Mishra *et al.*, 2019), compared with lower irrigation regimes have also been reported.

Table 1. Effect of irrigation schedule and sulphur fertilization on plant height (cm) and plant population m⁻² of Indian mustard

Treatments	Plant height (cm)		Plant population m ⁻²	
	30 DAS	60 DAS	At harvest	At harvest
Irrigation (I)				
I ₀ : Rainfed	24.98	114.26	145.06	30.67
I ₁ : Irrigation at 25-30 DAS (pre flowering)	25.60	120.76	159.08	32.00
I ₂ : Irrigation at flowering (F)	25.53	120.61	161.61	32.78
I ₃ : Irrigation at 25-30 DAS + F	26.72	129.58	172.66	32.44
I ₄ : Irrigation at 25-30 DAS + Siliquae formation (SF)	25.55	126.48	168.88	32.67
I ₅ : Irrigation at 25-30 DAS + F + SF	26.77	132.10	173.86	32.78
S.Ed±	0.62	1.74	2.24	1.02
CD: (P=0.05)	NS	3.87	4.99	NS
Sulphur (S)				
S ₀ : No application (Control)	25.02	114.53	152.94	31.78
S ₁ : 20 kg ha ⁻¹	26.13	127.13	166.87	32.28
S ₂ : 40 kg ha ⁻¹	26.43	130.24	170.75	32.61
S.Ed±	0.58	1.51	2.19	0.66
CD: (P=0.05)	NS	3.11	4.52	NS
Interaction (I x S)	NS	NS	NS	NS

NS- non significant, DAS-Days after sowing

Table 2. Effect of irrigation schedule and sulphur fertilization on dry matter accumulation of Indian mustard

Treatments	Dry matter accumulation (g plant ⁻¹)			
	25 DAS (Pre-flowering)	50 DAS (Flowering)	75 DAS (Siliquae formation)	At harvest
Irrigation (I)				
I ₀ : Rainfed	2.06	9.13	16.47	18.51
I ₁ : Irrigation at 25-30 DAS (pre flowering)	2.09	10.15	19.91	22.09
I ₂ : Irrigation at flowering (F)	2.11	9.98	21.96	24.23
I ₃ : Irrigation at 25-30 DAS + F	2.14	11.15	29.77	32.60
I ₄ : Irrigation at 25-30 DAS + Siliquae formation (SF)	2.12	10.57	28.86	31.42
I ₅ : Irrigation at 25-30 DAS + F + SF	2.14	11.22	30.04	32.84
S.Ed±	0.02	0.34	0.52	0.52
CD:(P=0.05)	NS	0.75	1.17	1.15
Sulphur (S)				
S ₀ : No application (Control)	2.09	9.27	19.99	22.04
S ₁ : 20 kg ha ⁻¹	2.11	10.69	26.36	28.97
S ₂ : 40 kg ha ⁻¹	2.13	11.15	27.16	29.84
S.Ed±	0.02	0.23	0.39	0.39
CD:(P=0.05)	NS	0.47	0.80	0.81
Interaction (I x S)	NS	NS	NS	NS

N.S: Non-significant, DAS-Days after sowing

Table 3. Effect of irrigation schedule and sulphur fertilization on number of branches of Indian mustard at harvest

Treatments	Number of branches plant ⁻¹		
	Primary branches plant ⁻¹	Secondary branches plant ⁻¹	Total branches plant ⁻¹
Irrigation (I)			
I ₀ : Rainfed	3.25	2.19	5.44
I ₁ : Irrigation at 25-30 DAS (pre flowering)	3.67	3.07	6.74
I ₂ : Irrigation at flowering (F)	3.77	3.29	7.03
I ₃ : Irrigation at 25-30 DAS + F	4.11	4.53	8.63
I ₄ : Irrigation at 25-30 DAS + Siliquae formation (SF)	4.08	4.47	8.56
I ₅ : Irrigation at 25-30 DAS + F + SF	4.19	4.81	9.01
S.Ed±	0.05	0.21	0.22
CD: (P=0.05)	0.11	0.48	0.49
Sulphur (S)			
S ₀ : No application (Control)	3.44	2.81	6.25
S ₁ : 20 kg ha ⁻¹	4.02	4.14	8.17
S ₂ : 40 kg ha ⁻¹	4.08	4.23	8.28
S.Ed±	0.06	0.10	0.11
CD: (P=0.05)	0.13	0.20	0.24
Interaction (I x S)	NS	NS	NS

N.S: Non-significant

Sulphur fertilisation significantly affected plant height (Table 1), dry matter accumulation (Table 2), number of branches plant⁻¹ (Table 3), and crop growth rate (Table 4), but not plant population at harvest. The highest plant height at 60 DAS and harvest, dry matter accumulation at 50 and 75 DAS, crop growth rate during 25-50 DAS, 50-75 DAS, and 75 DAS to harvest, and numbers of primary, secondary, and total branches plant⁻¹ were recorded with 40 kg S ha⁻¹ (S₂), followed by 20 kg S ha⁻¹ (S₁); the two levels were statistically at par and significantly higher than no sulphur fertilisation (S₀). Sulphur application enhances the nutrient absorption capacity of plants through its availability in the root zone, resulting in increased mustard plant height (S. Singh *et al.*, 2017). Similar results were reported by Dongarkar *et al.* (2005), A. K. Singh *et al.* (2017), and Rana *et al.* (2020).

Table 4. Effect of irrigation schedule and sulphur fertilization on crop growth rate of Indian mustard

Treatments	Crop growth rate (g m ⁻² day ⁻¹)			
	0-25 DAS	25-50 DAS	50-75 DAS	75 DAS- harvest
Irrigation (I)				
I ₀ : Rainfed	2.53	8.68	9.04	2.51
I ₁ : Irrigation at 25-30 DAS (pre flowering)	2.67	10.31	12.56	2.79
I ₂ : Irrigation at flowering (F)	2.77	10.31	15.77	2.98
I ₃ : Irrigation at 25-30 DAS + F	2.77	11.71	24.16	3.67
I ₄ : Irrigation at 25-30 DAS + Siliquae formation (SF)	2.76	11.06	23.95	3.35
I ₅ : Irrigation at 25-30 DAS + F + SF	2.80	11.90	24.69	3.67
S.E.d _±	0.10	0.49	1.06	0.09
CD: (P=0.05)	NS	1.10	2.37	0.21
Sulphur (S)				
S ₀ : No application(Control)	2.66	9.12	13.74	2.61
S ₁ : 20 kg ha ⁻¹	2.71	11.10	20.31	3.37
S ₂ : 40 kg ha ⁻¹	2.78	11.77	21.04	3.50
S.E.d _±	0.06	0.36	0.75	0.08
CD: (P=0.05)	NS	0.75	1.56	0.16
Interaction (I x S)	NS	NS	NS	NS

N.S: Non-significant, DAS-Days after sowing

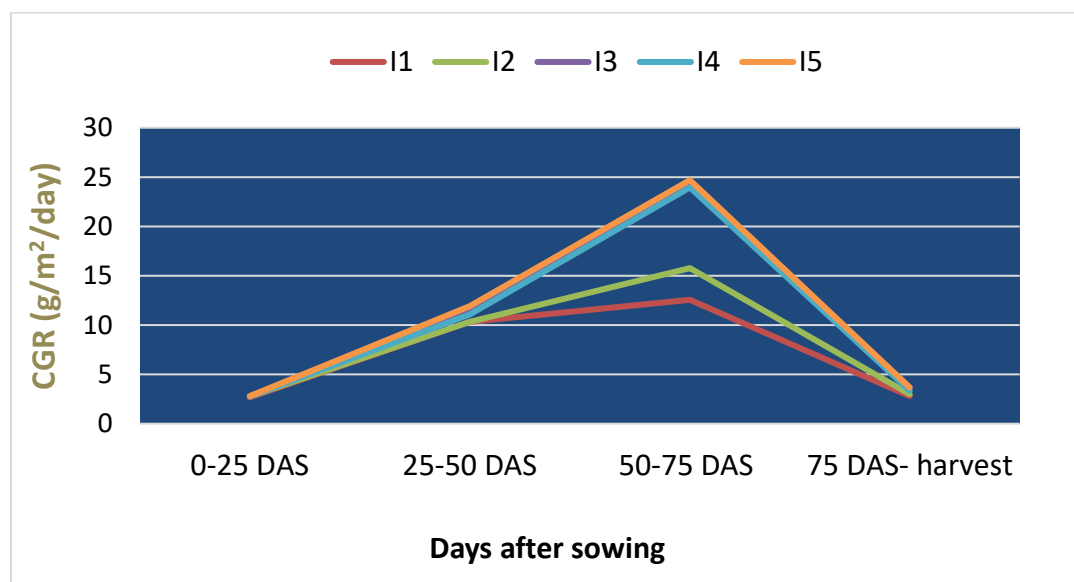


Fig. 1. Crop growth rate as influenced by irrigation schedule

3.2 Water Use and Water Use-efficiency

In the present investigation, the data (Table 5) revealed that three irrigations at 25-30 DAS + flowering + siliquae formation recorded the highest evapotranspiration and total water use by Indian mustard among the treatments. This was followed by two irrigations at 25-30 DAS + flowering and 25-30 DAS + siliquae formation, whereas the lowest evapotranspiration and total water use were recorded under rainfed conditions. However, crop and field water use efficiencies under irrigation schedules I₁ and I₀ were close to each other and comparatively higher than under the remaining treatments, while the lowest values were recorded with I₅. This may be because consumptive water use (CU) increases with irrigation frequency, whereas crop water use efficiency follows the opposite trend. Therefore, higher evapotranspiration and total water use under the higher irrigation regimes (two or three irrigations) were associated with comparatively lower crop and field water use efficiencies. Similar results in different crops have also been reported by Piri *et al.* (2011), Raut *et al.* (2000), and Sarma *et al.* (2007).

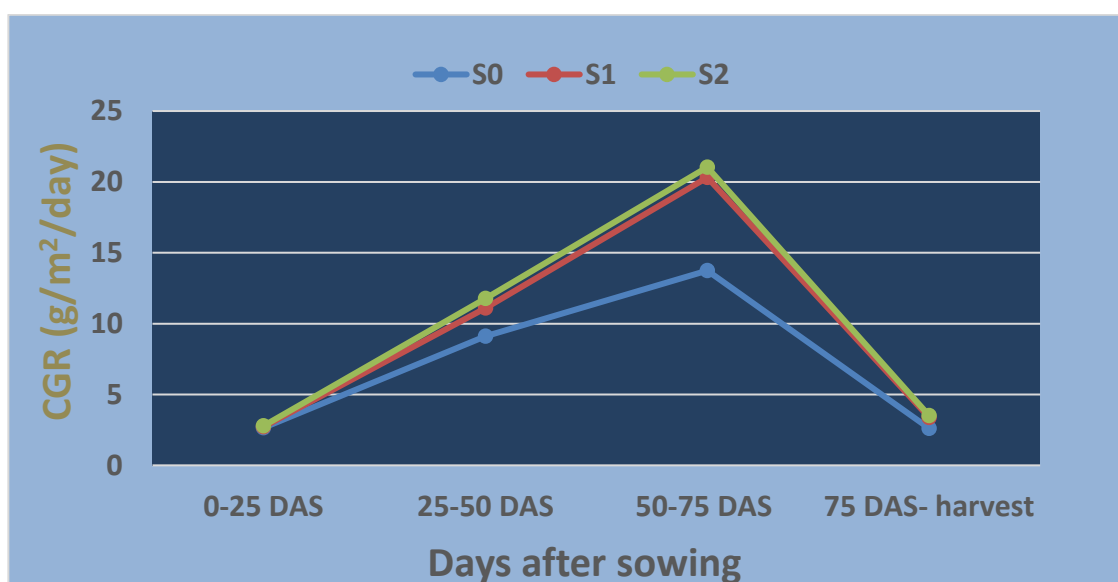


Fig. 2. Crop growth rate as influenced by sulphur fertilization

Table 5. Effect of irrigation schedule and sulphur fertilization on evapo-transpiration, total water use and water use efficiency of Indian mustard

Treatments	Evapo-transpiration (cm)	Total water use (cm)	Crop water use efficiency (kg ha ⁻¹ cm ⁻¹)	Field water use efficiency (kg ha ⁻¹ cm ⁻¹)
Irrigation (I)				
I ₀ : Rainfed	8.84	8.84	89.82	89.82
I ₁ : Irrigation at 25-30 DAS (pre flowering)	11.68	12.05	89.96	87.25
I ₂ : Irrigation at flowering (F)	12.14	12.62	88.54	85.12
I ₃ : Irrigation at 25-30 DAS + F	17.07	17.46	75.62	73.95
I ₄ : Irrigation at 25-30 DAS + Siliquae formation (SF)	16.38	16.90	78.78	76.30
I ₅ : Irrigation at 25-30 DAS + F + SF	21.69	22.66	60.75	58.17
Sulphur (S)				
S ₀ : No application (Control)	14.38	14.84	69.76	67.89
S ₁ : 20 kg ha ⁻¹	14.74	15.17	85.30	83.19
S ₂ : 40 kg ha ⁻¹	14.77	15.26	86.68	84.22

Table 6. Effect of irrigation schedule and sulphur fertilization on yield (q ha⁻¹), oil content and oil yield (q ha⁻¹) of Indian mustard

Treatments	Seed yield (kg ha ⁻¹)	Stover yield (kg ha ⁻¹)	Oil content (%)	Oil yield (kg ha ⁻¹)
Irrigation (I)				
I ₀ : Rainfed	795.52	1879.06	36.20	290.40
I ₁ : Irrigation at 25-30 DAS (pre flowering)	1052.97	2520.41	37.60	398.06
I ₂ : Irrigation at flowering (F)	1076.67	2591.78	38.03	413.29
I ₃ : Irrigation at 25-30 DAS + F	1291.55	3166.02	38.50	499.33
I ₄ : Irrigation at 25-30 DAS + Siliquae formation (SF)	1290.87	3145.12	38.22	496.07
I ₅ : Irrigation at 25-30 DAS + F + SF	1318.47	3244.74	38.69	511.85
S.Ed±	12.62	35.45	0.63	9.11
CD: (P=0.05)	28.12	78.99	1.41	20.29
Sulphur (S)				
S ₀ : No application (Control)	980.62	2301.78	34.82	342.58
S ₁ : 20 kg ha ⁻¹	1207.74	2958.45	39.11	474.31
S ₂ : 40 kg ha ⁻¹	1224.67	3013.34	39.69	487.61
S.Ed±	11.18	26.96	0.43	6.50
CD: (P=0.05)	23.07	55.65	0.88	13.41
Interaction (I x S)	NS	NS	NS	NS

N.S: Non-significant

Application of different sulphur levels resulted in little variation in evapotranspiration and total water use (Table 5). However, crop and field water use efficiencies increased with increasing sulphur levels. The highest crop water use efficiency was obtained with 40 kg S ha⁻¹, followed by 20 kg S ha⁻¹, both of which were considerably higher than the control. This may be due to the increase in seed yield resulting from sulphur application (Piri et al., 2011; Raut et al., 2000).

4. Conclusion

The study demonstrated clear main effects of irrigation scheduling and sulphur fertilisation on growth, water use, and productivity of Indian mustard under the irrigated rabi conditions of Assam. Three irrigations at 25–30 DAS, flowering, and siliquae formation produced the highest seed and oil yields, but required the greatest water use and resulted in the lowest crop and field water use efficiencies. Two irrigations, either at 25–30 DAS + flowering or at 25–30 DAS + siliquae formation, produced very similar seed yields (1291.55 and 1290.87 kg ha⁻¹, respectively) while using less water than the three-irrigation schedule. Sulphur at 40 kg ha⁻¹ produced the highest seed yield, oil content, and oil yield, although 20 kg ha⁻¹ was statistically at par with 40 kg ha⁻¹ for several growth traits. The reported irrigation × sulphur interactions were non-significant for the measured growth and yield variables. Overall, the findings indicate that critical-stage irrigation and adequate sulphur nutrition can improve Indian mustard productivity under the conditions of this study, with two irrigations providing a favourable balance between yield and water-use performance.

5. Limitations

The experiment was conducted at a single location in Jorhat during one rabi season (2019–20), using one Indian mustard variety and three replications. Consequently, the observed responses may be specific to the climatic and experimental conditions of that season and site. The irrigation treatments were evaluated only at the stated crop-growth stages and irrigation depth, so the findings apply to those management conditions. Multi-season and multi-location validation, including additional mustard genotypes, would strengthen the generalisability and confirm the stability of the observed yield and water-use responses.

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Declaration of AI Use

This manuscript was prepared through the intellectual contributions of the author(s) to the study design, data analysis, interpretation of results, and scholarly content. Grammarly and ChatGPT were used solely to assist with grammatical refinement and reference formatting during manuscript revision. The author(s) accept full responsibility for the accuracy, integrity, and final content of the manuscript.

Competing Interests

Authors have declared that they have no known competing financial interests or non-financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Baug, S., & Darvhankar, M. S. (2026). Climate change and drought challenges in Indian mustard (*Brassica juncea*) cultivation: Future prospects for sustainable production under climate-smart agriculture. *Plant Cell Biotechnology and Molecular Biology*, 27(3–4), 269–295. <https://doi.org/10.56557/pcbmb/2026/v27i3-410458>
- Dongarkar, K. P., Pawar, W. S., Khawale, V. S., Khutate, N. G., & Gudadhe, N. N. (2005). Effect of nitrogen and sulphur on growth and yield of mustard (*Brassica juncea* L.). *Journal of Soils and Crops*, 15(1), 163–167.
- Gupta, N. K., Sain, R. S., Gupta, D., Gupta, S., Singh, J., & Jain, N. K. (2024). Yield maximization in Indian mustard (*Brassica juncea* (L.) Czernj. & Cosson) through nutrient management and optimizing sowing date. *Journal of Plant Nutrition*, 47(17), 2824–2835. <https://doi.org/10.1080/01904167.2024.2369070>
- Hegde, D. M., & Sudhakara Babu, S. N. (2009). Declining factor productivity and improving nutrient-use efficiency in oilseeds. *Indian Journal of Agronomy*, 54(1), 1–8. ResearchGate article page
- Kumar, P., Kumar, S., Siddiqui, M. Z., Khan, N., Kumar, S., Kumar, A., Yadav, R., Bagri, J. P., & Pooja, K. (2025). Influence of irrigation scheduling and foliar application of phosphorus, sulphur, and boron on quality characteristic and soil nutrient availability of Indian mustard (*Brassica juncea* L.). *Archives of Current Research International*, 25(10), 292–304. <https://doi.org/10.9734/acri/2025/v25i101568>
- Mishra, J., Singh, R. K., Yadav, D., Das, A., & Sahoo, S. (2019). Effect of tillage and irrigation frequency on growth of Indian mustard [*Brassica juncea* (L.) Czernj and Cosson]. *International Journal of Chemical Studies*, 7(1), 2127–2130.
- Mohan, V. R., MacDonald, M. T., & Abbey, L. (2025). Impact of water deficit stress on Brassica crops: Growth and yield, physiological and biochemical responses. *Plants*, 14(13), 1942. <https://doi.org/10.3390/plants14131942>
- Mohapatra, A., Behura, A. K., Jena, J., Behera, S. D., Behera, B., & Prusty, A. A. (2024). Productivity and profitability of mustard [*Brassica juncea* (L.) Czern. and Coss.] under varying levels of sulphur and boron application methods. *Journal of Applied and Natural Science*, 16(3), 1062–1070. <https://doi.org/10.31018/jans.v16i3.5527>

- Piri, I., Nik, M. M., Tavassoli, A., Rastegaripour, F., & Babaeian, M. (2011). Effect of irrigation frequency and application levels of sulphur fertilizer on water use efficiency and yield of Indian mustard (*Brassica juncea*). *African Journal of Biotechnology*, 10(55), 11459–11467. <https://doi.org/10.5897/AJB.9000173>
- Rana, K., Parihar, M., Singh, J. P., & Singh, R. K. (2020). Effect of sulfur fertilization, varieties and irrigation scheduling on growth, yield, and heat utilization efficiency of Indian mustard (*Brassica juncea* L.). *Communications in Soil Science and Plant Analysis*, 51(2), 265–275. <https://doi.org/10.1080/00103624.2019.1705325>
- Raut, R. F., Hamid, A., Ganvir, M. M., Wanjari, S. S., & Hadole, S. S. (2000). Consumptive use, water use efficiency, seed yield and quality of mustard as influenced by irrigation and sulphur application. *Journal of Soils and Crops*, 10(2), 252–255.
- Ray, K., Pal, A. K., & Sengupta, K. (2014). *Effect of irrigation and sulphur on Indian mustard*. LAP Lambert Academic Publishing.
- Roy, K. C., Rahman, M. M., Rahman, K. A. M. M., Miah, M. A. M., Hossain, A., & Zaman, M. A. U. (2017). Effects of irrigation and sulphur application on growth, yield and oil content of rapeseed (*Brassica napus*). *Eco-friendly Agriculture Journal*, 10(3), 30–34.
- Sarma, A., Singh, H., & Nanwal, R. K. (2007). Effect of integrated nutrient management on productivity of wheat (*Triticum aestivum*) under limited and adequate irrigation supplies. *Indian Journal of Agronomy*, 52(2), 120–123. <https://doi.org/10.59797/ija.v52i2.4905>
- Shah, S. H., Parrey, Z. A., Islam, S., Tyagi, A., Ahmad, A., & Mohammad, F. (2022). Exogenously applied sulphur improves growth, photosynthetic efficiency, enzymatic activities, mineral nutrient contents, yield and quality of *Brassica juncea* L. *Sustainability*, 14(21), 14441. <https://doi.org/10.3390/su142114441>
- Sharma, S. K., Mendhe, S. N., Kolte, H. S., Rajput, G. R., & Yenpreddiwar, M. D. (2006). Effect of sowing and irrigation management on growth and yield of mustard (*Brassica juncea*). *Journal of Soils and Crops*, 16(2), 455–459.
- Singh, A. K., Meena, R. N., Kumar, A. R., Kumar, S., Meena, R., & Singh, A. P. (2017). Effect of land configuration methods and sulphur levels on growth, yield and economics of Indian mustard [*Brassica juncea* (L.)] under irrigated condition. *Journal of Oilseed Brassica*, 8(2), 151–157.
- Singh, S. P., Mahapatra, B. S., Kumar, A., Meena, V. S., & Joshi, N. (2025). Sustainable intensification of Indian mustard: Integrating irrigation, planting, and mulching for enhanced yield and soil health in the Shivalik Himalayas, India. *Environment, Development and Sustainability*. <https://doi.org/10.1007/s10668-025-06530-8>
- Singh, S., Thenua, O. V. S., & Singh, V. (2017). Effects of irrigation schedules and nutrient levels on mustard (*Brassica juncea* L.). *International Journal of Recent Advances in Multidisciplinary Research*, 4(11), 3056–3059.
- Yadav, R. D. (2005). *Effect of mulching and sulphur on mustard [Brassica juncea (L.) Czernj and Cosson] under varying levels of irrigation* [Doctoral dissertation, Rajasthan Agricultural University].

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